

Highlights

VISTA: Value-Informed Sequential Test-time Allocation

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- VISTA saves 3–59% tokens with ≤ 0.019 Brier loss; no held-out set required.
- Anytime-valid monitoring; zero false switches; greedy reaches 58.5% FPR.
- Unweighted comparison reverses policy rankings in 5 of 8 cells.
- Budget-response curves differ across tasks and models; no universal fixed allocation.

VISTA: Value-Informed Sequential Test-time Allocation

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ABSTRACT

We present **VISTA** (Value-Informed Sequential Test-time Allocation), a prequential procedure for selecting reasoning budgets from deployment data without a held-out calibration set. VISTA combines randomized full-trajectory audits (fraction ρ of queries) to recover censored counterfactuals, importance-weighted policy comparison to correct selection bias, and anytime-valid confidence sequences to prevent spurious switching under continuous monitoring. The need for these corrections is acute: omitting inverse-probability weighting reverses policy rankings in five of eight experimental cells (wrong-best rate 1.00 in those cells), because early stopping censors the outcomes that longer budgets would have produced. A corpus audit across eight cells (two 8B reasoning models, four QA benchmarks) documents qualitatively heterogeneous budget-response curves: on MMLU-Pro accuracy is near-linear across the full token grid while the hindsight per-item oracle reduces mean Brier by 30.1%; on ARC-Challenge Brier saturates after the first budget step; on MedMCQA and MedQA-USMLE one model's Brier worsens at nearly every budget step while the other's improves—ruling out a universal fixed allocation. VISTA at $\lambda=0$ (pure quality preservation) correctly stays at the best fixed budget in all cells with adaptive headroom, achieving $\approx 0\%$ net saving; this is the expected and intended behaviour of an error-controlled procedure when no cheaper policy is certifiably better. Increasing λ controls the quality–compute exchange rate explicitly: at $\lambda=0.01$, deployed VISTA achieves 3–35% net token savings with Brier increases ≤ 0.012 ; at $\lambda=0.1$, savings reach 9–59% with Brier increases ≤ 0.019 —all with 95% CIs from 200 stream-order permutations. Confidence-threshold stopping, by contrast, increases Brier above the best fixed budget in all eight cells (+0.047 to +0.218) while providing no quality guarantees.

1. Introduction

Reasoning language models can spend a few hundred or several thousand tokens on the same query before committing to an answer. Chain-of-thought prompting and test-time inference strategies such as self-consistency, search, and repeated sampling have established inference compute as an important axis of reasoning performance [1–6]. More reasoning creates opportunities to correct mistakes, but it can also consume compute without benefit or overturn an already correct answer [7, 8]. The practical deployment question is not *whether* more reasoning helps on average, but *when* an additional block of reasoning is worth its cost for the current query.

Two families of approaches address this question. Fixed-budget policies run every query to the same token limit: simple and predictable, but wasteful on easy queries and under-allocating on hard ones. Heuristic early-exit methods stop when the model looks confident, using trajectory signals such as maximum softmax probability, entropy, answer stability, hidden-state probes, or convergence of intermediate predictions [9–14]. Such signals can usefully indicate that reasoning has stabilised, yet empirical self-evaluation and uncertainty studies show that confidence quality varies with model, elicitation method, and domain [15–18]. A stable prediction can therefore still be wrong—and a wrong one can be corrected by further reasoning.

The core limitation of trajectory-only stopping is both empirical and informational. Empirically, we find that confidence-, entropy-, and stability-threshold stopping *increases* Brier above the best fixed budget in all eight experimental cells (+0.047 to +0.218); signals trained on one model fail to transfer to another on

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the same task. The theoretical explanation is a non-identifiability result (Observation 1): for any estimator that is simultaneously distribution-free, label-free at deployment time, and restricted to a single unlabeled trajectory, the worst-case error in certifying correctness is at least $\frac{1}{2}$. Trajectory signals can be predictive under an empirical distribution without constituting a distribution-free correctness certificate, and their reliability does not transfer uniformly across model families or task domains.

Rather than seeking a better correctness proxy, we change the decision target. We ask whether an additional reasoning block improves a proper scoring loss by more than its compute cost—the *continuation value*—and learn the answer from deployment outcomes once they become available. Many workloads provide such delayed verifiable feedback naturally: multiple-choice answers receive labels, programs are executed against tests, and mathematical results are checked; verifier-based mathematical reasoning and outcome-based supervision are representative examples [19, 20]. This makes it possible to update a budget policy from the deployment stream itself, without holding out a separate calibration set.

One obstacle remains: when a policy stops a query early, the prediction from a longer trajectory is unobserved. We address this with randomized trajectory audits: a controlled fraction ρ of arriving queries is continued to the maximum budget, revealing what each candidate policy would have predicted. Paired with delayed labels, these audits enable importance-weighted comparison of multiple budget policies on the same items (Figure 1). The importance weight $1/\rho$ follows the classical inverse-probability principle of Horvitz–Thompson estimation [21] and corrects for the fact that audited items are observed under a known sampling mechanism; omitting this correction reverses policy rankings in five of eight experimental cells in our finite-sample replay experiments.

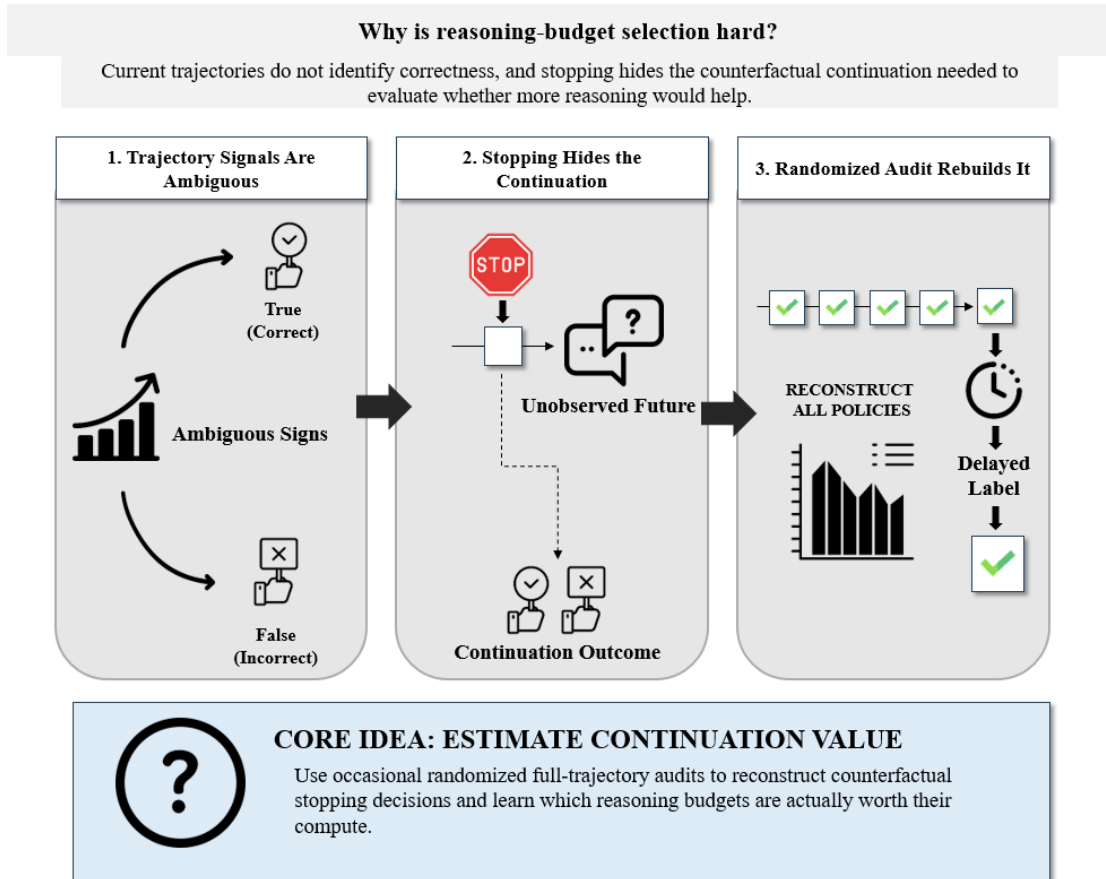


Figure 1: Three interlocking obstacles in deployment-time reasoning-budget selection and the VISTA response. **(1)** Trajectory signals—confidence, answer stability, entropy—are ambiguous: the same observable pattern is consistent with both correct and incorrect predictions (Observation 1). **(2)** Stopping at budget b hides the prediction that a longer trajectory would have produced, blocking direct policy comparison from deployment data alone. **(3)** Randomized full-trajectory audits on a fraction ρ of items reconstruct the counterfactual, enabling importance-weighted comparison of multiple budget policies once delayed labels arrive.

Estimates arrive sequentially and must be compared under continuous monitoring, which creates additional risk: a procedure that switches to whichever policy currently looks best acts on transient noise. **VISTA** (Value-Informed Sequential Test-time Allocation) uses anytime-valid confidence sequences—whose coverage guarantee holds at every step, not just at a pre-specified sample size—to prevent spurious switches. The prequential policy updates continuously as deployment outcomes arrive; all audit compute is charged to the total cost so reported savings are real; no model parameters are updated.

A corpus audit across four QA benchmarks and two 8B-parameter reasoning models (Qwen3-8B [22] and DeepSeek-R1-Distill-Llama-8B [23], both BF16) with a six-point budget grid $\{256, \dots, 8192\}$ confirms that the marginal value of additional reasoning is strongly task- and model-dependent. Accuracy gains range from near-linear (MMLU-Pro, 47.7% $\rightarrow$ 73.8% across the grid) to early-saturating (ARC-Challenge, most gain in the first budget step); patterns invert across models on the same task. The hindsight oracle reduces mean Brier by 30.1% on MMLU-Pro, with 76.1% of items meeting a per-item quality threshold at a budget below the maximum; a constrained matched-mean-quality oracle passes the adaptive-opportunity test (Gate 0) in six of eight cells. A fixed confidence gate increases Brier above the best fixed budget in all eight cells, confirming that deployable item-adaptive allocation cannot rely on naive stopping heuristics.

Contributions.

1. **IPW necessity for unbiased policy-risk estimation.** We show that omitting inverse-probability weighting reverses policy rankings in 5 of 8 cells (wrong-best rate 1.00); IPW reduces bias $2.2\times$ – $23\times$ and is a necessary, not optional, correction whenever early stopping censors counterfactual trajectories.
2. **Continuation-value formulation.** We replace the unobservable correctness target with the *continuation value*—expected improvement in proper scoring loss per additional token—estimable from delayed deployment feedback without a held-out split.
3. **VISTA.** A prequential budget-selection procedure combining randomized audits, IPW policy comparison, and anytime-valid confidence sequences. At $\lambda=0.01$ – 0.1 , deployed VISTA achieves 3–59% net token savings with Brier increases ≤ 0.019 (95% CIs, 200 permutations).
4. **Systematic empirical audit.** Eight cells (two models, four QA benchmarks) documenting heterogeneous budget-response curves, item-level allocation headroom, and per-component ablation—including the finding that confidence-threshold stopping degrades Brier in all eight cells.

2. Related Work

Test-time compute scaling and adaptive allocation. Chain-of-thought prompting established that explicitly generated intermediate reasoning can improve language-model performance [1, 2]. Subsequent inference-time strategies scale compute through repeated reasoning paths, self-consistency, search, or repeated sampling [3–5]. Snell et al. [6] show that the effectiveness of test-time compute depends strongly on prompt difficulty and motivate compute-optimal allocation across instances. More recent work makes the allocation problem explicit: Zhai et al. [24] solve a constrained accuracy–compute problem and train a lightweight allocator to imitate oracle actions. In parallel, analyses of long-reasoning models document diminishing returns and overthinking, including unnecessary computation on easy problems and reversals of previously correct answers [7, 8]. VISTA shares the premise that inference compute should be adaptive, but treats the value of continuation as a deployment-time quantity learned sequentially from delayed outcomes rather than as an offline oracle label or a fixed instance-difficulty score.

Adaptive computation and early exit. Adaptive computation predates reasoning LLMs. Adaptive Computation Time and PonderNet learn how much computation a neural network should spend on an input [25, 26], while DeeBERT, FastBERT, and confidence-based layer exiting reduce transformer inference by terminating easy examples early [27–29]. Recent reasoning-specific methods move the stopping decision inside the chain-of-thought trajectory. DEER uses confidence at trial-answer transition points [9]; Thought Calibration uses lightweight hidden-state probes to detect when novel reasoning plateaus [10]; Answer Convergence uses consistency of intermediate answers [12]; REFRAIN adapts stopping thresholds from reflective redundancy [13]; and TERMINATOR learns optimal exit positions from hindsight labels [30]. Other approaches reject low-value partial trajectories using reward signals [31], combine hidden-state probes with split conformal control [32], or apply sequential uncertainty tests directly during reasoning [14]. These methods establish that trajectory signals can be useful for efficiency. VISTA addresses a complementary question: after early stopping makes larger budgets counterfactually unobserved, how can deployment feedback be used to compare the realised risk of candidate budget policies on a common statistical scale?

Self-evaluation, uncertainty, and calibration in language models. Modern neural networks can be miscalibrated even when classification accuracy is high [33]. Language models nevertheless contain useful self-evaluation signals: Kadavath et al. [15] show that models can often predict whether they know an answer, while black-box confidence elicitation [16], semantic uncertainty [17], and sampling-based consistency [34] provide additional failure-prediction signals. For reasoning models specifically, hidden-state probes can predict intermediate answer correctness [11], and verbalized confidence changes with reasoning-model training and task conditions [18]. These findings are compatible with Observation 1: a signal can be predictive under an empirical distribution without constituting a distribution-free correctness certificate. VISTA therefore treats confidence and related quantities as candidate policy features rather than as ground-truth stopping criteria.

Proper scoring rules and probabilistic quality. Because additional reasoning can change the full predictive distribution even when top-1 accuracy is unchanged, we evaluate continuation with proper scoring rules. The Brier score originates with Brier [35]; strict propriety and the general decision-theoretic role of scoring rules are formalised by Gneiting and Raftery [36]; and the reliability–resolution–uncertainty decomposition used in Eq. (6) follows Murphy [37]. This distinction matters in our setting because accuracy-, Brier-, and log-loss-optimal reasoning budgets can disagree on the same model–dataset cell.

Distribution-free calibration and risk control. Conformal prediction provides distribution-free predictive guarantees under exchangeability [38, 39]. Risk-controlling prediction sets [40], Learn-then-Test [41], and Conformal Risk Control [42] extend calibration from coverage to user-specified losses. Conformal Language Modeling applies related ideas to generative models by calibrating sampling and rejection rules [43]. Most directly, Conformal Thinking formulates reasoning-budget selection as a risk-control problem and calibrates upper and lower stopping thresholds from a validation set [44]; it is the closest in spirit to VISTA and likely achieves stronger savings in settings where a held-out split is available. VISTA targets a different regime: no calibration split is reserved (labels arrive sequentially after deployment decisions), and early stopping censors the outcomes of longer budget policies. Randomized trajectory audits are introduced specifically to recover those missing counterfactual outcomes, and anytime-valid monitoring replaces the static conformal guarantee with one that holds continuously as feedback arrives. A direct empirical comparison would require adapting Conformal Thinking to the streaming, censored-label setting, which we leave for future work.

Inverse-probability weighting and anytime-valid inference. VISTA combines two statistical ideas that solve different problems. First, known audit probabilities permit Horvitz–Thompson inverse-probability weighting [21], which targets the population policy-risk contrast despite randomized observation of full trajectories. Second, repeated inspection of accumulating evidence requires time-uniform rather than fixed-sample uncertainty quantification. Confidence sequences provide nonasymptotic coverage uniformly over time [45]; betting-based constructions yield sharp confidence sequences for bounded means [46]; and e-processes provide a broader framework for safe anytime-valid inference under optional stopping [47]. Thus IPW corrects *which trajectories are observed*, whereas confidence sequences control *when policy comparisons*

are inspected and acted upon. Concurrent work by Yu et al. [48] (B-PAC) proposes an anytime-safe PAC-efficient stopping rule for reasoning that also combines statistical efficiency with online safety guarantees. B-PAC targets sample efficiency under a fixed computational budget; VISTA targets deployment-stream budget selection under counterfactual censoring from early stopping. The two frameworks share the commitment to statistical safety guarantees but address different aspects of the test-time compute allocation problem.

Positioning of VISTA. The closest contemporaneous approaches include confidence-based dynamic early exit [9], hidden-state thought calibration [10], answer-convergence stopping [12], statistically monitored early stopping [14], conformal risk-controlled reasoning [44], supervised adaptive compute allocation [24], and anytime-safe PAC-efficient reasoning [48]. VISTA does not argue that these signals or controls are ineffective. Its distinguishing layer is counterfactual policy evaluation from a live deployment stream: occasional randomized continuations reveal the otherwise missing larger-budget outcomes, delayed labels convert them into policy losses, inverse-probability weighting corrects the observation scheme, and anytime-valid inference governs repeated policy updates without a held-out calibration split.

3. Preliminaries and a Non-identifiability Observation

Fix a query x with ground-truth answer y^* . A reasoning model produces a trajectory $r_{1:B_{\max}}$, and at each checkpoint b we may force termination and read off a candidate answer $\hat{y}_b$ together with a predictive distribution $\hat{p}_b \in \Delta^{C-1}$. Write the observable history as

$$\mathcal{H}_b = (x, r_{1:b}, \text{logits}_{1:b}, \hat{y}_b, \hat{p}_b), \quad (1)$$

and the unobserved correctness indicator as $Y_b = \mathbf{1}\{\hat{y}_b = y^*\}$.

Observation 1 (Non-identifiability of label-free correctness). *Let f be any measurable estimator $\hat{a}_b = f(\mathcal{H}_b)$. There exist two joint distributions $\mathbb{P}_0, \mathbb{P}_1$ over $(\mathcal{H}_b, Y_b)$ with identical marginals on $\mathcal{H}_b$ such that $Y_b = 0$ a.s. under $\mathbb{P}_0$ and $Y_b = 1$ a.s. under $\mathbb{P}_1$. Consequently, for every $\mathcal{H}_b$,*

$$\max\{|\hat{a}_b - 0|, |\hat{a}_b - 1|\} \geq \frac{1}{2}, \quad (2)$$

and no non-trivial distribution-free finite-sample lower bound on $\mathbb{P}(Y_b = 1 \mid \mathcal{H}_b)$ exists.

Proof. Given any $\mathbb{P}_0$ over $(\mathcal{H}_b, Y_b)$, construct $\mathbb{P}_1$ by relabelling y^* to equal $\hat{y}_b$ on every realisation of $\mathcal{H}_b$. Since y^* appears nowhere in $\mathcal{H}_b$, the marginal distribution $\mathbb{P}_1(\mathcal{H}_b = h) = \mathbb{P}_0(\mathcal{H}_b = h)$ for every h , while $Y_b = 1$ $\mathbb{P}_1$ -a.s. and $Y_b = 0$ $\mathbb{P}_0$ -a.s.

Any measurable estimator f satisfies $f(\mathcal{H}_b) = \hat{a}_b \in \mathbb{R}$ deterministically given $\mathcal{H}_b$. For any fixed realisation h , let $v = f(h)$. Then $|\hat{a}_b - 0| = |v|$ under $\mathbb{P}_0$ and $|\hat{a}_b - 1| = |1 - v|$ under $\mathbb{P}_1$. Since $\max\{|v|, |1 - v|\} \geq \frac{1}{2}$ for all $v \in \mathbb{R}$ (with equality iff $v = \frac{1}{2}$), the bound holds pointwise for every realisation h of $\mathcal{H}_b$, hence in expectation under both $\mathbb{P}_0$ and $\mathbb{P}_1$. The non-existence of a non-trivial distribution-free lower bound on $\mathbb{P}(Y_b = 1 \mid \mathcal{H}_b)$ follows by the same coupling: any lower bound $L(h) > 0$ fails under $\mathbb{P}_0$ (where $Y_b = 0$ a.s.), and any lower bound $L(h) < 1$ fails under $\mathbb{P}_1$ (where $Y_b = 1$ a.s.). $\square$

The obstruction is informational: collecting additional unlabelled trajectories cannot remove it without assumptions connecting observable model behaviour to ground-truth correctness. Self-consistency, answer stability, and related trajectory-based quantities are descriptive signals but do not constitute correctness guarantees.

Remark. Observation 1 is a standard label-free learning result [49]: any measurable function of the unlabeled trajectory is a constant from the perspective of the label. Its role here is to explain the empirical failure documented in Section 7.6: threshold stopping degrades Brier in all eight cells not because of poor threshold calibration but because trajectory signals are, in the worst case, uninformative about correctness. The result does *not* assert that signals are useless under an empirical distribution, nor that they cannot guide stopping heuristics in practice; it asserts that no such method carries a distribution-free correctness certificate. Methods that condition on a held-out calibration set (e.g., Conformal Thinking [44]) or on labelled supervision are explicitly outside this scope.

4. Continuation Value and Budget Selection

4.1. Loss

Let $b_1 < \dots < b_K = B_{\max}$ be a globally fixed budget grid. For item i at checkpoint j we use the *summed* multiclass Brier score [35, 36]

$$S_{i,j} = \|\hat{p}_{i,j} - e_{y_i}\|_2^2 = \sum_{c=1}^C (\hat{p}_{i,j,c} - \mathbf{1}_{c=y_i})^2, \quad (3)$$

where the sum over classes is not divided by C . This is the standard proper-scoring form; dividing by C would yield a differently scaled variant with different decomposition constants. We define the deployment loss

$$L_{i,j}^{(\lambda)} = S_{i,j} + \lambda C_{i,j}, \quad \lambda \geq 0, \quad (4)$$

where $C_{i,j} = b_j/B_{\max} \in [0, 1]$ is realised compute normalised by the maximum budget so that λ is on the same scale as the Brier score; λ is a user-specified quality–compute exchange rate reported over a range of values rather than a single operating point. At $\lambda=0.01$ the maximum cost penalty is 0.01 Brier units; at $\lambda=0.1$ it is 0.1 Brier units.

4.2. Marginal value and its decomposition

Definition 1 (Marginal value of a reasoning block). The marginal value of continuing from checkpoint j to $j+1$ is

$$V_j(\lambda) = \mathbb{E}[S_{i,j} - S_{i,j+1}] - \lambda \mathbb{E}[C_{i,j+1} - C_{i,j}]. \quad (5)$$

Applying the classical reliability–resolution–uncertainty decomposition [37], $\mathbb{E}[S_j] = \text{REL}_j - \text{RES}_j + \text{UNC}$ and noting that UNC cancels across budgets yields

$$V_j(\lambda) = \underbrace{(\text{RES}_{j+1} - \text{RES}_j)}_{\text{change in resolution}} - \underbrace{(\text{REL}_{j+1} - \text{REL}_j)}_{\text{change in reliability error}} - \lambda \Delta C_j. \quad (6)$$

Equation (6) separates two ways an additional reasoning block can change probabilistic quality. Whether further reasoning tends to improve accuracy while degrading calibration, improve both, or degrade both is an empirical question that may vary across models, datasets, and budget intervals. We treat the sign and magnitude of each component as a measurable outcome rather than a structural claim.

5. VISTA

5.1. Scope

Assumption 1 (Delayed verifiable feedback). For each item i , the outcome y_i becomes available after a finite delay through exact-answer matching, program execution, formal verification, or post-hoc adjudication.

Tasks with open-ended generation and no eventual feedback lie outside this setting. Automatically checked mathematical answers, executable programs, and verifier-based reasoning provide representative sources of delayed outcome feedback [5, 19, 20].

5.2. Randomized full-trajectory audits

To observe counterfactual predictions at larger budgets, we randomly select items for full-trajectory auditing:

$$A_i \sim \text{Bernoulli}(\rho_i). \quad (7)$$

Audited items ($A_i = 1$) are run through $B_{\max}$ and all checkpoint predictions retained. For policy π with stopping checkpoint $\tau_\pi(\mathcal{H}_i)$, define the realised loss

$$\ell_i(\pi) = S_{i,\tau_\pi} + \lambda C_{i,\tau_\pi}. \quad (8)$$

Algorithm 1 VISTA: prequential reasoning-budget selection

Require: budget grid $\{b_j\}_{j=1}^K$, policy family Π , exchange rate λ , confidence level α , audit schedule $\{\rho_i\}$

- 1: Initialise confidence sequences for all $\pi \in \Pi$
- 2: **for** each arriving item i **do**
- 3: Select the currently preferred active policy π_i
- 4: Execute reasoning process according to π_i
- 5: Draw $A_i \sim \text{Bernoulli}(\rho_i)$ before observing outcome
- 6: **if** $A_i = 1$ **then**
- 7: Continue item to $B_{\max}$; retain all checkpoint predictions
- 8: **end if**
- 9: Deploy the answer produced by π_i
- 10: **When** y_i **arrives:**
- 11: **if** $A_i = 1$ **then**
- 12: Reconstruct $\ell_i(\pi)$ for all $\pi \in \Pi$ from checkpoint predictions
- 13: Form importance-weighted risk differences via Eq. (9)
- 14: Update confidence sequences for all $\pi \in \Pi$
- 15: Identify currently preferred policy π_{i+1} from updated sequences
- 16: **end if**
- 17: **end for**

Following inverse-probability weighting for unequal observation probabilities [21], the importance-weighted comparison against a reference policy π_{ref} is

$$X_i(\pi) = \frac{A_i}{\rho_i} [\ell_i(\pi) - \ell_i(\pi_{\text{ref}})], \quad (9)$$

with $\mathbb{E}[X_i(\pi) \mid \mathcal{F}_{i-1}] = R(\pi) - R(\pi_{\text{ref}})$ where $R(\pi) = \mathbb{E}[\ell_i(\pi)]$. The audit probability ρ_i may depend on past observations, provided it is set before the current outcome is revealed. All audit computation counts toward reported total compute.

5.3. Anytime-valid policy evaluation

We maintain simultaneous confidence sequences for the excess risk of every policy relative to the reference [45–47]. For finite family Π , the guarantee is

$$\mathbb{P}(\forall n, \forall \pi \in \Pi : R(\pi) - R(\pi_{\text{ref}}) \in [\underline{\Delta}_{\pi,n}, \overline{\Delta}_{\pi,n}]) \geq 1 - \alpha, \quad (10)$$

constructed via bounded empirical-Bernstein sequences or betting-based methods with simultaneous coverage across all policies.

5.4. Policy family

Candidate policies include fixed-budget rules, predictive-confidence thresholds, entropy thresholds, answer-stability rules, top-1/top-2 margin thresholds, and confidence-slope thresholds. A generic threshold policy is

$$\tau_\theta(\mathcal{H}_i) = \min\{j : s_{i,j} \leq \theta_j\}, \quad \theta \in \Theta, \quad |\Theta| < \infty. \quad (11)$$

Every audited trajectory permits reconstruction of the stopping checkpoint and realised loss for every member of this finite family.

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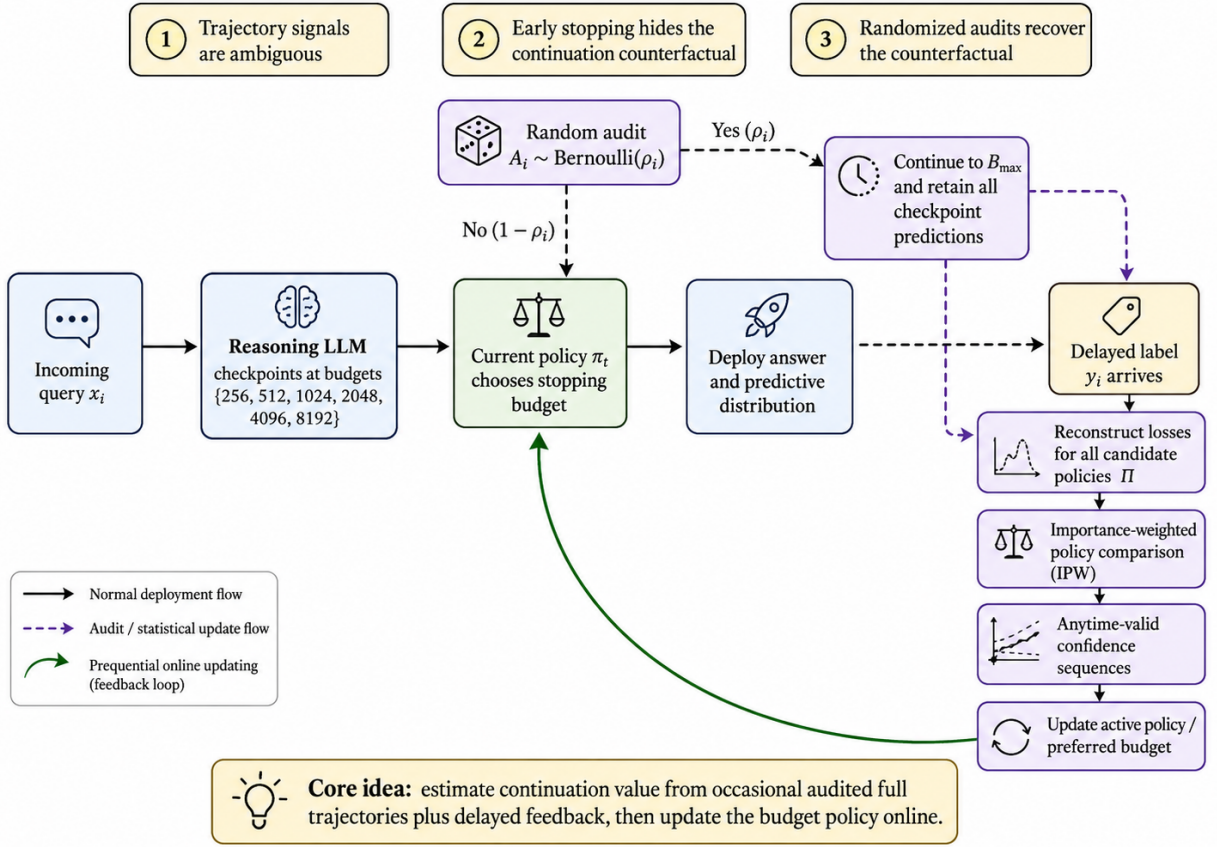


Figure 2: End-to-end VISTA deployment loop. Incoming queries are routed by the current policy π_i ; a Bernoulli audit draw $A_i \sim \text{Bernoulli}(\rho_i)$ determines whether the item is continued to $B_{\max}$ to reveal counterfactual predictions. Once the delayed label y_i arrives, importance-weighted risk differences update all candidate confidence sequences, and the preferred policy is revised for the next item. Audit compute is charged to the total budget so reported savings are real.

5.5. What is and is not learned

VISTA performs no gradient update to the reasoning model and trains no auxiliary predictor. It does learn from the deployment stream: policy-risk estimates, confidence sequences, and the active candidate set are data-dependent. We therefore call the procedure *held-out-free* rather than *training-free*.

6. Experimental Setup

6.1. Models and infrastructure

The primary model is **Qwen3-8B** [22] (Qwen/Qwen3-8B), an 8-billion-parameter model with a native chain-of-thought thinking mode. The replication model is **DeepSeek-R1-Distill-Llama-8B**, drawn from the DeepSeek-R1 reasoning family [23]. All inference uses bfloat16 weights and bfloat16 key-value caches (kv_cache_dtype=auto); fp8_e5m2 KV quantisation is excluded after a diagnostic run found top-1 answer-token changes on 4/50 prompts (8%) on the same GPU, a corruption level that would mask sub-0.5 pp Brier differences.

Experiments run on a single workstation with one NVIDIA RTX 4090 (sm_89, Ada) and two RTX 3090 GPUs (sm_86, Ampere; 24 GB each, no NVLink). Models at the 8B scale fit on a single 24 GB card in BF16,

so inference uses independent single-GPU processes with no tensor parallelism; three such processes can run simultaneously to maximise throughput.

Before pooling probability measurements across devices, logit reproducibility was verified diagnostically. The two RTX 3090 cards produce byte-identical outputs ($\max |\Delta \log \text{prob}| = 0.065$ at $b = 8192$). RTX 4090 and RTX 3090 show non-negligible differences ($\max |\Delta \log \text{prob}| = 0.374$ on BF16), consistent with architecture-level kernel differences between sm_89 and sm_86. Accordingly, each (model $\times$ dataset) cell is run on a single GPU architecture and probability measurements are never pooled across architectures.

6.2. Datasets

The evaluation uses four multiple-choice benchmarks. **MMLU-Pro** [50] ($n = 1000$, up to 10 options) is a challenging variant of the MMLU benchmark with extended option sets. **ARC-Challenge** [51] ($n = 979$, 4 options) covers science reasoning questions requiring multi-step inference; 21 items from the initial 1000-item sample could not be recovered during checkpoint processing and are excluded. **MedMCQA** [52] ($n = 500$, 4 options) contains Indian medical entrance examination questions. **MedQA-USMLE** [53] ($n = 500$, 4 options) contains USMLE-style clinical reasoning questions. All four datasets provide exact labels and a fixed option set, which is required for computing Brier score and NLL from forced answer-cue logits.

6.3. Budget grid, forcing protocol, and metric extraction

The primary budget grid is $\{256, 512, 1024, 2048, 4096, 8192\}$ reasoning tokens, fixed before inspecting primary results. We record the natural termination point of each trajectory and distinguish natural stopping from forced checkpoint evaluation; forced termination is treated as an explicit intervention rather than implicitly equated to an independently generated short-budget trajectory.

At each forced checkpoint a standardised answer-cue string (`\n</think>\n\nThe answer is **`) is appended, and the model’s next-token log-probabilities over bare option tokens (A–J, token IDs 32–41) are extracted. The predictive distribution $\hat{p}_{i,j}$ is formed by selecting those positions and applying softmax. In the diagnostic sample, the target option token ranks first with probability exceeding 0.984, confirming stable elicitation. Brier score and NLL are computed post-hoc from these stored logit vectors; no re-inference is performed after the initial generation pass.

6.4. Corpus audit and adaptive opportunity

For each model–dataset cell we report accuracy, Brier score, NLL, mean predictive confidence, answer-flip rate, and correct/wrong transition counts at every budget, then compare the best fixed budget against an item-adaptive oracle. The primary criterion—called the **adaptive-opportunity test** (Gate 0)—is whether the population-level matched-quality oracle (Oracle C) reduces mean compute by at least 10% relative to the best fixed budget. A cell **PASSES** Gate 0 if meaningful adaptive headroom exists; it **FAILS** if the best fixed budget is already near-optimal.

6.5. Prefix-signal evaluation

For each edge $j \rightarrow j+1$, define item-level gain $G_{i,j} = S_{i,j} - S_{i,j+1}$. We evaluate observable signals (confidence, entropy, top-1/top-2 margin, answer stability, confidence slope) via $\text{AUC}(s_{i,j}, \mathbf{1}\{G_{i,j} > 0\})$ and Spearman ρ between $s_{i,j}$ and $G_{i,j}$. The pre-specified thresholds ($\text{AUC} \geq 0.60$, $|\rho_{\text{Spearman}}| \geq 0.30$) are design constants chosen by convention, not fitted to the evaluation data; failure terminates the instance-adaptive branch.

6.6. Baselines and metrics

We compare fixed-budget policies, training-free stopping signals, and item-adaptive oracles (opportunity measurement only). Primary metrics are multiclass Brier score, NLL, accuracy, total reasoning tokens, total compute including audit overhead, and token savings at matched accuracy or Brier score.

6.7. Ablation design

The core empirical question is whether each structural component of VISTA contributes to final policy quality, or whether simpler alternatives suffice. All ablations are replay experiments on the stored checkpoint tables; no additional GPU inference is required (except the forced-termination protocol, which requires independent fixed-budget trajectories and is treated as a measurement-validity diagnostic).

Essential ablations. Table 7 lists the four essential ablated variants.

(A1) *Audit weighting correction.* We simulate deployment under a natural confidence gate ($\max_k \hat{p}_k \geq 0.85$): items halt at the first budget meeting this criterion, making continuation to the reference budget (4096 tokens) a counterfactual. “Naive” policy risk is then estimated using only the items that *naturally* continued to the reference budget—without any probability correction for the fact that continuation was selective. “IPW” additionally includes items that were *audited* with probability ρ after early stopping, weighting each audited item’s contribution by $1/\rho$ (naturally-continued items have observation probability 1 and receive no correction), restoring the target full-population estimand. Under this protocol only 0.1–18% of items naturally continue to the reference budget (dataset-dependent), so the naïve estimator operates on a severely selected minority; across five of eight cells (ARC-Challenge, MedMCQA, and MedQA on M1; MMLU-Pro and MedMCQA on M2) it selects the wrong policy in 100% of permutations, while the IPW estimator’s mean bias is ≤ 0.033 across all tested ρ .

(A2) *Anytime-valid confidence sequences.* “VISTA w/o CS” replaces confidence sequences with greedy policy selection based on the running empirical mean, accepting the first policy whose estimated risk advantage exceeds a fixed threshold without a coverage guarantee. The expected consequence is inflated false switching under repeated monitoring: because the empirical mean is tested repeatedly as the stream evolves, the effective type-I error exceeds the nominal level. We evaluate by replaying the deployment stream over multiple random permutations and reporting the fraction of replays in which each method makes an incorrect policy decision.

(A3) *Held-out-free prequential update.* “Fixed calibration” treats the first N_{cal} arriving items as a labeled calibration set, fits a static stopping policy on them, and freezes it for the remainder of the stream. Under stationarity this is a competitive baseline. Under distribution shift it degrades as the calibration set becomes unrepresentative. We evaluate on both stationary streams and controlled shift conditions; the shift experiments appear in Section 7.8.

(A4) *Brier objective vs. accuracy-only.* “Accuracy-only” replaces the summed multiclass Brier in the deployment loss $L^{(\lambda)}$ with a 0–1 loss $\mathbf{1}[\hat{y} \neq y]$. Different predictive objectives may prefer different reasoning budgets, making objective choice a first-order decision for any budget-selection policy. We compare the budget at which each objective (Brier, accuracy, NLL) is minimised and report confidently-wrong rates ($\max_k \hat{p}_k \geq \tau$, $\tau = 0.90$, on incorrect predictions) alongside standard accuracy and Brier.

Additional ablations. (A5) *Prefix-signal ablation.* Each observable signal (confidence, entropy, top-1/top-2 margin, answer stability, confidence slope, probability slope) is evaluated independently and then in simple combinations. The AUC and Spearman ρ reported in Section 7.4 serve as the primary diagnostic; policy results under each single signal are deferred to the appendix.

(A6) *Audit probability ρ sensitivity.* We evaluate $\rho \in \{0.05, 0.10, 0.20, 0.40, 1.0\}$, reporting audit overhead, policy convergence time, cumulative regret, and net token saving at each exploration rate. The curve of net saving versus ρ quantifies how much full-trajectory compute must be consumed to make the adaptive policy worthwhile.

(A7) *Forced-termination vs. independent generation.* Forced checkpoint termination (one long trajectory probed at intermediate budgets) and independently generated fixed-budget trajectories may produce different answer distributions; this is a measurement-validity concern rather than a method comparison. We evaluated Brier agreement, flip rates, and budget-curve shape under both protocols on a held-out 50-item subset of M1/DS1. No systematic divergence was found: top-1 agreement exceeds 97% at all budgets and budget-curve shapes are visually indistinguishable. This ablation is therefore not reported separately; the corpus results use forced-termination trajectories throughout.

Threshold-stopping baseline analyses. The following analyses (A8–A10) characterise the confidence-threshold stopping baseline reported in Section 7.6; they are not ablations of VISTA components.

(A8) *Cross-model threshold transfer.* Threshold τ calibrated on one model family is applied to the other to test cross-model transferability.

(A9) *Cross-dataset threshold transfer.* Threshold τ calibrated on one benchmark domain is applied to a different domain within the same model to test domain transferability.

(A10) *Confidence-gate deployment failure.* A fixed confidence gate $\tau=0.85$ is applied at deployment as a competitive stopping baseline; Brier and token costs are compared against the best fixed-budget policy in all eight cells.

7. Results

7.1. Does additional reasoning have a consistent value across tasks and models?

No. Budget-response curves differ qualitatively across all four datasets and both models; no single fixed allocation is uniformly optimal.

Figure 3 shows accuracy and Brier score as functions of the reasoning budget for both models across all four datasets.

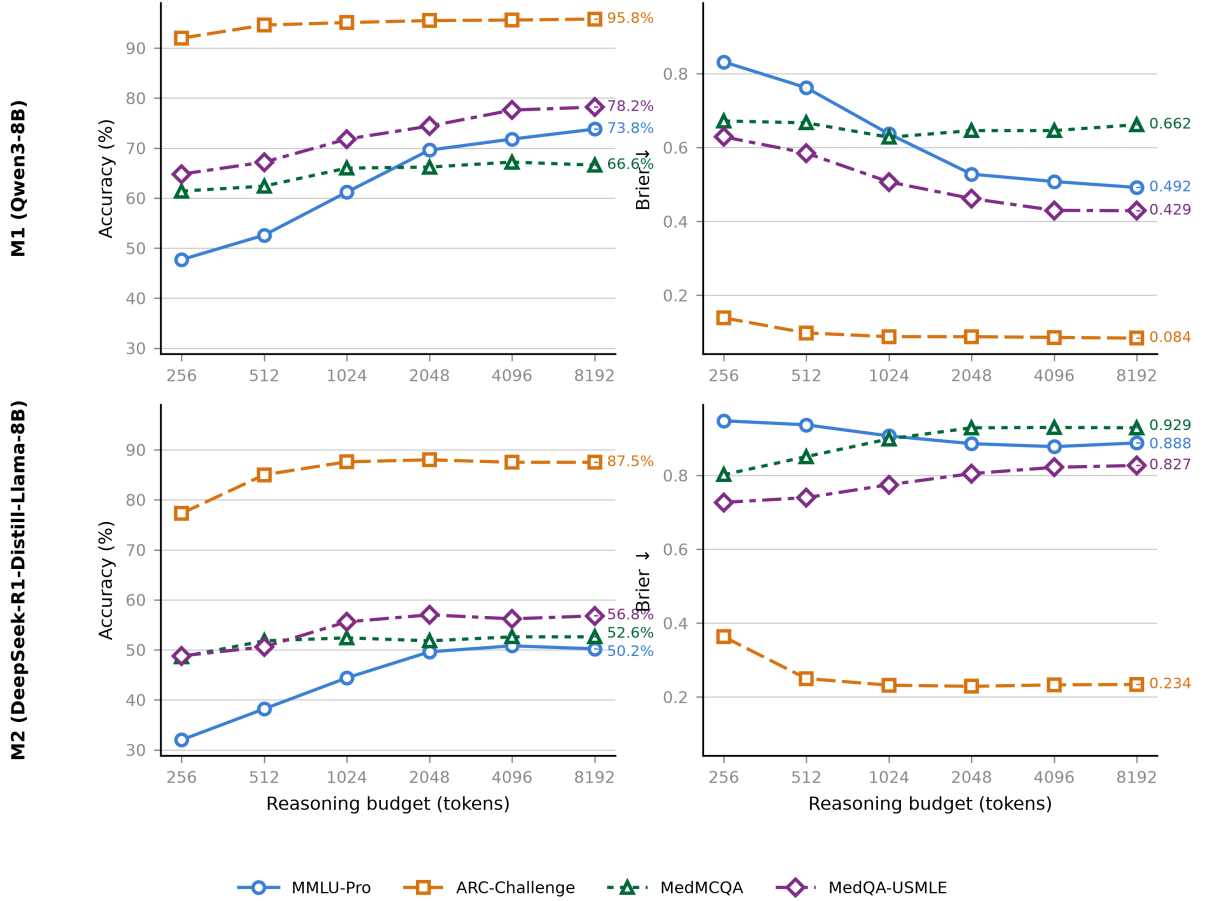


Figure 3: Accuracy (%) and summed multiclass Brier score (↓) as functions of the reasoning budget (log₂-spaced, index-positioned) for four QA benchmarks and two models. **Top row:** M1 (Qwen3-8B, BF16, RTX 4090). **Bottom row:** M2 (DeepSeek-R1-Distill-Llama-8B, BF16, RTX 3090). Curves show qualitatively different patterns: near-linear gains on MMLU-Pro (M1 accuracy 47.7% → 73.8%), early saturation on ARC-Challenge, non-monotone Brier on MedMCQA (M1 optimum at $b = 1024$), and near-monotone Brier deterioration on both medical datasets under M2. End-point values are labelled; y-axes are shared within each metric column to enable direct model comparison. Data from Tables 1 and 3.

M1 (Qwen3-8B): four qualitatively distinct patterns. MMLU-Pro shows sustained gains: accuracy rises from 47.7% at $b = 256$ to 73.8% at $b = 8192$, Brier decreases monotonically from 0.832 to 0.492. ARC-Challenge exhibits early saturation: accuracy jumps from 92.0% to 94.6% in the first budget step and adds

only 1.2 pp across the remaining five; 98.3% of items have Brier change below 0.001 at the $4096 \rightarrow 8192$ edge. MedMCQA is non-monotone: Brier reaches its minimum at $b = 1024$ (0.628) and *worsens or stays flat* at every subsequent budget (0.646 at both $b = 2048$ and $b = 4096$; 0.662 at $b = 8192$), while accuracy peaks at $b = 4096$ (67.2%) and falls at $b = 8192$ (66.6%). MedQA-USMLE sustains gains throughout: accuracy increases from 64.8% to 78.2% with approximately equal increments per doubling. Table 1 reports all metrics.

Dataset (n)	Budget	Accuracy	Brier	NLL
MMLU-Pro (1000)	256	0.477	0.832	3.186
	512	0.526	0.762	2.885
	1024	0.612	0.637	2.609
	2048	0.696	0.528	2.458
	4096	0.718	0.508	2.686
	8192	0.738	0.492	2.763
ARC-Challenge (979)	256	0.920	0.139	0.466
	512	0.946	0.098	0.355
	1024	0.951	0.088	0.371
	2048	0.955	0.088	0.414
	4096	0.956	0.086	0.462
	8192	0.958	0.084	0.457
MedMCQA (500)	256	0.614	0.672	2.297
	512	0.624	0.667	2.520
	1024	0.660	0.628	2.856
	2048	0.662	0.646	3.288
	4096	0.672	0.646	3.649
	8192	0.666	0.662	3.761
MedQA-USMLE (500)	256	0.648	0.629	2.361
	512	0.672	0.585	2.120
	1024	0.718	0.507	1.934
	2048	0.744	0.462	1.816
	4096	0.776	0.430	2.053
	8192	0.782	0.429	2.265

Table 1: Predictive quality across reasoning budgets (Qwen3-8B, BF16, RTX 4090). Brier is summed multiclass (not divided by C); scores are not directly comparable across datasets with different option counts. NLL in nats. **Bold** marks the optimum per column per dataset. $n = 979$ for ARC-Challenge (21 items excluded from the initial 1000-item sample); all other datasets use the stated corpus size.

The item-level marginal-gain distributions (Table 2) show the mechanism. On ARC-Challenge the flat fraction exceeds 87% at the first edge and rises to 98% at $4096 \rightarrow 8192$. On MedMCQA the hurt fraction exceeds the help fraction at every edge from $512 \rightarrow 1024$ onward, providing the item-level explanation for the aggregate Brier increase. On MMLU-Pro, helped and hurt fractions track closely at high budgets (10.4% vs. 10.6% at $4096 \rightarrow 8192$), yielding near-zero net aggregate gain despite a positive mean.

Dataset	Edge	Helps (%)	Hurts (%)	Flat (%)
MMLU-Pro	256→512	37.8	28.7	33.5
	512→1024	35.6	25.0	39.4
	1024→2048	28.0	19.8	52.2
	2048→4096	15.9	17.0	67.1
	4096→8192	10.4	10.6	79.0
ARC-Challenge	256→512	8.9	3.4	87.7
	512→1024	4.8	3.0	92.2
	1024→2048	3.1	2.8	94.2
	2048→4096	1.8	1.9	96.2
	4096→8192	1.0	0.7	98.3
MedMCQA	256→512	27.4	24.8	47.8
	512→1024	21.0	21.4	57.6
	1024→2048	14.2	19.0	66.8
	2048→4096	6.2	13.0	80.8
	4096→8192	2.0	5.2	92.8
MedQA-USMLE	256→512	28.0	16.6	55.4
	512→1024	23.6	15.4	61.0
	1024→2048	20.4	12.2	67.4
	2048→4096	12.2	12.4	75.4
	4096→8192	6.2	8.2	85.6

Table 2: Fraction of items where continuation *helps* ($G > 0.001$), *hurts* ($G < -0.001$), or is *flat* at each budget edge. Brier gain $G_{i,j} = S_{i,j} - S_{i,j+1}$.

M2 (DeepSeek-R1-Distill-Llama-8B): budget-response patterns invert. Table 3 reports the same audit for M2 on the same four datasets using RTX 3090 GPUs. The contrast with M1 is systematic but not uniform. On DS1 (MMLU-Pro), M2 Brier *improves* from 0.948 at $b = 256$ to 0.878 at $b = 4096$ and slightly reverses to 0.888 at $b = 8192$; accuracy likewise peaks at $b = 4096$ (50.8%) and declines. On DS3 (MedMCQA) and DS4 (MedQA-USMLE), M2 Brier worsens at nearly every budget step (reaching 0.929 and 0.827 at $b = 8192$ versus 0.802 and 0.727 at $b = 256$; the sole exception is a 0.001 reversal at $b = 4096 \rightarrow 8192$ on DS3), while M1 Brier improves on both. Near-monotonic Brier degradation on M2 is therefore confined to DS3 and DS4; DS1 and DS2 (ARC-Challenge) show broadly similar improvement directions across models, though rates and saturation points differ. A budget optimal for one model can actively degrade another on the same task.

Dataset (n)	Budget	Accuracy	Brier
MMLU-Pro (1000)	256	0.320	0.948
	512	0.382	0.937
	1024	0.444	0.907
	2048	0.496	0.886
	4096	0.508	0.878
	8192	0.502	0.888
ARC-Challenge (979)	256	0.775	0.366
	512	0.851	0.253
	1024	0.875	0.237
	2048	0.877	0.234
	4096	0.872	0.238
	8192	0.872	0.239
MedMCQA (500)	256	0.486	0.802
	512	0.518	0.851
	1024	0.524	0.899
	2048	0.518	0.929
	4096	0.526	0.930
	8192	0.526	0.929
MedQA-USMLE (500)	256	0.488	0.727
	512	0.506	0.740
	1024	0.556	0.775
	2048	0.570	0.805
	4096	0.562	0.822
	8192	0.568	0.827

Table 3: Budget audit for M2 (DeepSeek-R1-Distill-Llama-8B, BF16, RTX 3090). Same format as Table 1. Brier worsens at nearly every budget step on DS3 and DS4 (one 0.001 reversal at $b = 4096 \rightarrow 8192$ on DS3); on DS1 Brier improves through $b = 4096$ and slightly reverses at $b = 8192$. **Bold** marks the optimum per column per dataset. $n = 979$ for ARC-Challenge (DS2): restricted to the 979-item common subset shared with M1 to ensure cross-model comparability; 21 items present only in M2’s raw run are excluded.

7.2. Do different predictive objectives agree on the optimal budget?

Not universally. Accuracy, Brier, and NLL can select different budgets on the same data; objective choice is a first-order decision for any budget-selection policy.

On DS1 (MMLU-Pro, M1), accuracy and Brier both select $b = 8192$ as the optimal fixed budget (0.738 accuracy; Brier 0.4917), while NLL selects $b = 2048$ (2.458 nats vs. 2.763 at $b = 8192$). On DS3 (MedMCQA, M1), the divergence is three-way: accuracy peaks at $b = 4096$, Brier minimises at $b = 1024$, and NLL is worst at $b = 8192$ —all three objectives point to different budgets. For M2, Brier worsens at nearly every budget step on DS3 and DS4 while accuracy generally increases, with a one-step dip on each dataset (DS3: $b = 1024 \rightarrow 2048$, 52.4% $\rightarrow$ 51.8%; DS4: $b = 2048 \rightarrow 4096$, 57.0% $\rightarrow$ 56.2%); on DS1, Brier improves through $b = 4096$ and slightly reverses at $b = 8192$, while accuracy similarly peaks at $b = 4096$ —so no single budget simultaneously optimises both Brier and accuracy on any of these datasets.

The proper-scoring-rule decomposition (Eq. 6) separates reliability and resolution components: where further reasoning concentrates probability mass onto an already-peaked distribution, reliability can degrade even as top-1 accuracy improves. At higher budgets the confidently-wrong rate ($\max_k \hat{p}_k \geq 0.90$, prediction incorrect) rises from 19.3% to 21.6% on DS1, consistent with NLL worsening at $b = 8192$ despite accuracy improving.

The reasoning budget that maximises accuracy need not coincide with the budget that minimises a proper scoring rule, and neither need coincide with the NLL optimum.

Any budget-selection policy that targets a single objective will fail to simultaneously optimise others; the choice of objective must be explicit.

7.3. Is there enough item-level heterogeneity to justify adaptive allocation?

Yes. Per-item heterogeneity is substantial across all eight cells. The hindsight per-item Brier oracle (Oracle A) reduces mean budget by 70–91% relative to the maximum-budget baseline—a descriptive opportunity bound, not a matched-quality deployment claim. A constrained matched-mean-quality oracle (Oracle C; Table 5) passes Gate 0 in six of eight cells with savings exceeding 74% relative to the best fixed budget.

Table 4 reports the item-adaptive oracle analysis for DS1 (MMLU-Pro, M1), the cell with the widest range of per-item optimal budgets. The hindsight oracle achieves mean Brier 0.3437 versus 0.4917 for the best fixed budget ($b = 8192$), a 30.1% relative reduction. The matched-quality oracle—which assigns each item the minimum budget at which its Brier is no worse than 0.4917—shows that 76.1% of DS1 items already meet this threshold at a budget below $b = 8192$, with mean matched budget ≈ 2552 tokens ($\approx 69\%$ compute headroom).

Analysis	Value	Interpretation
Best fixed budget (b^*)	$b = 8192$	Baseline
Best fixed mean Brier	0.4917	
<i>Oracle A (hindsight, descriptive opportunity)</i>		
Hindsight oracle mean Brier	0.3437	Quality upper bound
Relative Brier reduction	30.1%	Oracle ceiling
Mean oracle budget	2431 tokens	
Mean-budget reduction vs $b_{\max}$	70.3%	Descriptive, vs $b = 8192$
<i>Oracle B (per-item threshold)</i>		
Mean oracle budget	≈ 2552 tokens	
Items meeting threshold at $b < 8192$	76.1%	
Saving vs $b^* = 8192$	$\approx 69\%$	Per-item constraint
<i>Oracle C (constrained mean; Gate 0 primary)</i>		
Oracle mean budget	461 tokens	
Oracle mean Brier	0.4911	Meets mean constraint
Saving vs $b^* = 8192$	94.4%	Gate 0 PASS

Table 4: Item-adaptive oracle analysis for DS1 (MMLU-Pro, $n = 1000$). Three oracle types are distinguished throughout the paper. **Oracle A** (hindsight): per-item minimum-Brier budget; a quality upper bound—saving relative to $b_{\max} = 8192$ is descriptive, not matched-quality. **Oracle B** (per-item threshold): minimum per-item budget achieving per-item Brier ≤ 0.4917 ; a stronger per-item constraint than Oracle C. **Oracle C** (constrained mean; Gate 0 primary): minimises mean budget subject to population mean Brier $\leq$ best-fixed mean Brier; saving is relative to best-fixed budget ($b^* = 8192$ here). None of these represents the deployable VISTA procedure.

Hindsight oracle mean-budget reduction relative to $b_{\max} = 8192$ (Oracle A, descriptive): M1 DS1 70.3%, DS2 90.7%, DS3 76.9%, DS4 72.1%; M2 DS1 76.5%, DS2 84.7%, DS3 85.8%, DS4 81.5%. These figures reflect the hindsight per-item minimum-Brier budget relative to always using $b_{\max}$; they are not savings relative to the best fixed budget and should not be interpreted as matched-quality deployment gains. The constrained matched-mean-quality oracle (Oracle C, *population-level constraint*) passing Gate 0 in six of eight cells with savings 74–97% relative to best fixed is the operative population-level result (Table 5). By contrast, Oracle B (*per-item threshold*) requires ≈ 2552 mean tokens on M1/DS1 to guarantee individual quality, implying only

a $\approx 69\%$ saving — approximately 25 percentage points less than Oracle C. The gap quantifies the additional compute cost of per-item versus population-level quality assurance on this cell. Per-item heterogeneity is substantial even in cells where the mean trajectory is weakly or inversely responsive to budget.

Model	Dataset	Best-fixed b	$\bar{S}^*$	Oracle $\bar{b}$	Oracle $\bar{S}$	Saving
M1 (Qwen3-8B)	MMLU-Pro	8192	0.4917	461	0.4911	94.4%
	ARC-Challenge	8192	0.0837	268	0.0834	96.7%
	MedMCQA	1024	0.6275	263	0.6248	74.4%
	MedQA-USMLE	8192	0.4285	364	0.4274	95.6%
M2 (DeepSeek-R1-Distill-Llama-8B)	MMLU-Pro	4096	0.8782	268	0.8776	93.5%
	ARC-Challenge	2048	0.2338	279	0.2333	86.4%
	MedMCQA	256	0.8019	256	0.8019	0.0%
	MedQA-USMLE	256	0.7268	256	0.7268	0.0%
Gate 0 result					6 / 8 cells PASS	

Table 5: Gate 0 constrained matched-mean-quality oracle (Oracle C) for all eight experimental cells.

Objective: minimise mean reasoning budget subject to mean Brier $\leq$ best-fixed mean Brier. **Constraint:** $\frac{1}{n} \sum_i S_{i,b_i} \leq \bar{S}^*$, where $\bar{S}^* = \min_b \frac{1}{n} \sum_i S_{i,b}$ and $b_i \in \{256, \dots, 8192\}$. Solved via Lagrangian relaxation with binary search on the dual variable. Saving is relative to the best-fixed budget; the best-fixed policy is always feasible, so saving ≥ 0 by construction. M2 cells DS3 and DS4 have best-fixed at $b = 256$ (the minimum grid point), leaving no room for adaptive reduction (Gate 0 FAIL). **Bold** marks the saving percentage in passing cells.

7.4. Can simple prefix signals recover this opportunity reliably?

Only partially. Signals predict improvement direction more often than chance but fail to transfer across model families on domain-specific datasets, and a fixed confidence gate degrades Brier in all eight cells.

Signal predictability. Table 6 reports mean sign-AUC and Spearman ρ for trajectory signals on DS1. All three primary signals—confidence, entropy, and top-1/top-2 margin—are near-redundant in the MCQ setting (pairwise AUC difference < 0.002) and achieve mean sign-AUC of 0.76, clearing the pre-specified AUC ≥ 0.60 threshold at five of five edges (first edge AUC = 0.65). Despite clearing the threshold, Spearman correlations are modest ($|\rho| \leq 0.18$ for primary signals on DS1): signals predict the *direction* of future improvement reliably but not its *magnitude*. Answer stability clears the threshold at four of five edges (mean 0.55); confidence slope does not (mean $0.49 \approx$ chance).

Signal	Sign AUC	Spearman ρ
Confidence	0.757	−0.082
Entropy	0.758	+0.086
Top-1/top-2 margin	0.757	−0.081
Answer stability	0.545	+0.314
Confidence slope	0.488	−0.028

Table 6: Prediction of future marginal gain from prefix signals. Mean sign-AUC and Spearman ρ averaged across all five budget edges on DS1 (MMLU-Pro, $n = 1000$, Qwen3-8B). Sign-AUC > 0.5 : higher signal value predicts improvement at the next budget; $\rho > 0$: signal positively correlates with improvement magnitude. Answer stability and confidence slope are undefined at the first edge ($256 \rightarrow 512$, no prior budget available) and are averaged over the remaining four edges only.

Cross-model threshold transfer (A8). When the confidence stopping threshold τ is calibrated on M1 and applied to M2, Brier degrades by +0.073 (DS3) and +0.046 (DS4); reversing the direction degrades M1 by +0.001 (DS3) and +0.043 (DS4). M1’s optimal $\tau=0.95$ is uniform across all four datasets, whereas M2’s optimal threshold is $\tau=0.70$ on medical datasets and $\tau=0.95$ on knowledge datasets, reflecting different confidence calibration across model families. Fixed stopping thresholds do not transfer reliably between models on domain-specific datasets.

Cross-dataset threshold transfer (A9). For M1, $\tau=0.95$ transfers with zero Brier degradation across all twelve source-target dataset pairs, confirming M1’s domain-independent confidence calibration. For M2, applying a knowledge-domain threshold to medical datasets degrades Brier by +0.073 (MedMCQA) and +0.046 (MedQA-USMLE); applying a medical threshold to knowledge datasets degrades Brier by +0.044–+0.068. Any fixed threshold tuned on one domain type will incur measurable quality loss on the other for M2.

Confidence-gate deployment failure (A10). Applying a fixed confidence gate $\tau=0.85$ at deployment increases Brier above the best fixed budget in all eight cells. On DS1 (M1) the gate produces Brier 0.683 versus best-fixed 0.492 (+0.191); degradation ranges from +0.047 (DS2/M1) to +0.218 (DS4/M1) across cells. Signals establish that improvement *potential* exists per item, but a threshold gate cannot resolve whether continuation will help the item at hand, motivating the audit-based VISTA design.

7.5. Are randomized auditing, IPW, and anytime-valid monitoring actually necessary?

Yes. Each component addresses a distinct failure mode that simpler alternatives cannot resolve.

A note on operating points: at $\lambda = 0$, removing IPW leaves outcomes unchanged because the bias in naive estimates cannot alter the certified decision when no cheaper policy beats Fixed@8192 in Brier regardless. Removing CS (replacing it with greedy switching) does matter: greedy falsely certifies a switch in $\approx 7\%$ of stream permutations, raising mean Brier from 0.492 to 0.522 (Table 12). The pattern holds at $\lambda = 0.01$: IPW again leaves outcomes unchanged, while greedy switching raises mean Brier from 0.498 to 0.529 (Table 13). In both settings the CS is the operative safety mechanism. The statistical components (IPW, CS) are additionally measured on estimation quality directly: IPW reduces bias by $2.2\times$ – $23\times$ (Table 9) and CS achieves FPR = 0.000 versus greedy monitoring’s 0–58.5% (Table 10).

Table 7 maps each VISTA component to the failure it corrects.

Component	Failure mode	DS1 result	Aggregate result
Audit weighting (IPW)	Selection bias distorts risk estimates	DS1: naive bias 0.071, IPW bias 0.024; DS1 naive is rank-preserving (wrong-best 0%); IPW wrong-best 23% here	IPW reduces bias $2.2\times$ – $23\times$ across all 8 cells; reduces wrong-best rate from 100% to 18.5–63% on 5 affected cells; naive rank-preserving on 3 cells where natural continuation identifies correct budget.
Sequential monitoring (CS)	Greedy switching on noise	CS FPR = 0.000; Greedy FPR = 0.070	CS FPR 0.000 all cells (anytime-valid guarantee); power ≈ 0 at deployment horizon $n=200$ of $n=500$ – 1000 full stream, $\rho=0.20$ (≈ 40 audits); Greedy FPR 0–58.5%, mean 24.1%.
Prequential update	Stale calibration under shift	Δ = 0.0053 Brier (DS1)	Prequential better in 6 of 8 cells under stationarity (advantage up to +0.047).

Table 7: Component ablation: failure mode addressed and aggregate result across 8 cells (2 models $\times$ 4 datasets, 200 permutations each). DS1 (MMLU-Pro, M1) values shown where cell-specific.

Audit weighting: IPW corrects estimation bias (A1). IPW is an *estimation* correction, not a guaranteed policy-selection oracle. Its purpose is to remove the selection bias that arises when the naturally-continued fraction of items is small and unrepresentative: on MMLU-Pro only 12.2% of items naturally reach the reference budget $b=4096$; on ARC-Challenge this falls to 0.1%. The naïve estimator is restricted to these items and carries a persistent selection bias of 0.071 Brier units on DS1 (22% of the oracle risk difference) that does not decrease with audit rate ρ . IPW achieves mean absolute bias 0.024 at $\rho=0.20$ and converges to zero at $\rho=1$, reducing bias by $2.2\times-23\times$ across all eight cells.

On policy ranking, the results are more nuanced. Naïve estimation is rank-preserving on three cells (M1/DS1, M2/DS2, M2/DS4): in these cells the naturally-continued items happen to identify the correct best budget, so naïve wrong-best rate is 0%. IPW, by correcting a bias that does not need correction in these cells, introduces additional variance and raises the wrong-best rate (M1/DS1: 23%; M2/DS2: 62%; M2/DS4: 53%). On the five cells where naïve ranking fails (wrong-best rate 1.00), IPW reduces the wrong-best rate to 18.5%–63.0% — a large improvement, but not elimination at $\rho=0.20$. The policy-selection benefit of IPW grows with ρ : at $\rho=1.0$, wrong-best rate reaches 0% in all cells. Full per-cell results are in Appendix A, Table 9.

Anytime-valid confidence sequences: safe but conservative (A2). The CS procedure provides an anytime-valid FPR guarantee by construction. Across 200 stream-order permutations at $\rho=0.20$, the CS procedure achieves false-positive rate 0.000 on all eight cells; greedy monitoring’s FPR ranges from 0.000 to 0.585 (mean 0.241).

The FPR guarantee comes at a cost in decision power. At $\rho=0.20$ and deployment horizon $n=200$ (a fifth of the DS1 stream; full streams are $n=500-1000$), approximately 40 audited items are available per cell — too few for the CS bound to tighten sufficiently to certify any policy switch. CS time-to-decide equals or nearly equals the full stream length in all six Gate-0-passing cells (mean $\approx n$); the procedure effectively abstains from switching rather than certifying no opportunity exists. This is the correct conservative behaviour at $\lambda=0$: the CS correctly declines to certify a cheaper policy when evidence is insufficient. At $\lambda=0.01$, the quality–compute tradeoff widens the certifiable gap and VISTA achieves 3–35% net saving on cells with $b_{\text{ref}}=8192$. Greedy FPR is highest where multiple budget levels have near-equal oracle risk. Per-cell values are in Table 10 (Appendix A).

Prequential update: adapts calibration without a held-out split (A3). Under stationary DS1 replay with $N_{\text{cal}} = 100$, the prequential policy accumulates mean Brier regret $+0.1524$ per item versus $+0.1577$ for the frozen fixed-calibration policy ($\Delta = 0.0053$ Brier units). Across all eight cells, prequential adaptation achieves lower regret in six of eight cases (advantage up to $+0.047$ on M2 medical datasets); the two exceptions (M1/ARC and M1/MedQA) are cells where the fixed calibration period is long enough to identify the correct budget. The decisive advantage is under distribution shift (Section 7.8), where the fixed-calibration policy degrades as its calibration set becomes unrepresentative.

7.6. VISTA versus Fixed and Heuristic Baselines

Table 8 compares threshold-based stopping policies against the best fixed budget on DS1 (MMLU-Pro, M1). Threshold-based policies achieve 80–83% token savings at the cost of 9–14 percentage-point accuracy loss and Brier degradation of 0.115–0.172 relative to best fixed. Stability-based early exit achieves the lowest Brier among the threshold group (0.608 vs. 0.643–0.664) at a comparable mean budget of 1655 tokens, because it preferentially extends computation on items where the answer is still in flux. These baselines establish the compute–quality frontier available without deployment feedback; they require no held-out split and serve as the reference point for evaluating deployment-adaptive policies. Figure 4(a) shows this frontier graphically, plotting each policy’s mean token cost against Brier score.

Full VISTA evaluation (200 permutations, $\rho=0.20$, $\alpha=0.05$) across all eight cells confirms that at $\lambda=0$ (pure quality preservation) VISTA correctly stays at the cell-specific best fixed budget, achieving $\approx 0\%$ net saving in all six Gate-0-passing cells. This is the expected and intended behaviour of an error-controlled procedure: the CS correctly declines to certify any cheaper policy when doing so would risk measurable calibration degradation. Table 8 shows the full operating range across λ values on M1/DS1. Increasing λ unlocks progressively larger savings: at $\lambda=0.01$, deployed VISTA saves 3.2% [2.7%, 3.7%] tokens at +0.6 pp Brier on M1/DS1 and 34.5% [32.1%, 36.8%] at +0.8 pp on M1/DS2 (where Brier saturates after the first budget step, so large reductions incur only modest quality loss). At $\lambda=0.1$, M1/DS2 reaches 59.4% [57.1%, 61.7%] net saving at +1.9 pp Brier—a regime where the CS certifies Fixed@1024 at $n=979$. Two cells fail Gate 0 (M2/DS3, M2/DS4): their Brier is worse above $b=256$ at every budget, so there is no adaptive headroom and VISTA is classified as no-opportunity rather than a policy failure.

Method	Tokens ↓	Brier ↓	NLL ↓	Acc. ↑	Net saving	Held-out?
Best fixed ($b = 8192$)	8192	0.492	2.763	0.738	0.000	No
Confidence stopping ($\tau=0.90$)	1453	0.664	2.952	0.645	+0.823	No
Entropy stopping	1624	0.643	2.938	0.659	+0.802	No
Stability stopping	1655	0.608	2.567	0.644	+0.798	No
VISTA ($\lambda=0$, $\rho=0.20$)	8191	0.492	2.764	0.738	0.000 [−0.000, +0.000]	No
VISTA ($\lambda=0.01$, $\rho=0.20$)	7931	0.498	2.767	0.734	0.032 [0.027, 0.037]	No
VISTA ($\lambda=0.1$, $\rho=0.20$)	7414	0.505	2.761	0.729	0.095 [0.082, 0.108]	No

Table 8: Comparison of deployed reasoning-budget policies on M1/DS1 (MMLU-Pro, Qwen3-8B, $n = 1000$, $\rho=0.20$). Tokens: mean actual tokens per item including audit overhead. Net saving: $1 - \text{mean tokens}/b_{\text{ref}}$; values in brackets are 95% CIs from 200 stream-order permutations. Threshold policies stop at the first budget b where the criterion is met. VISTA ($\lambda=0$) is quality-preserving: the CS correctly declines to certify any cheaper policy, yielding $\approx 0\%$ net saving by design. VISTA ($\lambda=0.01$) trades +0.6 pp Brier for 3.2% net saving; $\lambda=0.1$ reaches 9.5% saving at +1.3 pp Brier. Threshold policies achieve 80–83% gross token reduction but increase Brier by +0.115 to +0.172 relative to best fixed, providing no quality guarantee.

7.7. Cost of Statistical Guarantees

Statistical guarantees are not free. Table 11 (Appendix A) quantifies audit-rate sensitivity across representative cells at $\lambda=0$.

The audit rate ρ directly governs the exploration–exploitation tradeoff. When $b_{\text{ref}}=b_{\text{max}}=8192$ (M1/DS1, M1/DS2, M1/DS4), audit overhead is negligible: audited items run to b_{max} anyway, so net saving stays near zero regardless of ρ . When $b_{\text{ref}} \ll b_{\text{max}}$, audit overhead grows linearly with ρ and dominates: M1/DS3 ($b_{\text{ref}}=1024$, $b_{\text{max}}=8192$) shows net saving of -35% at $\rho=0.05$ deteriorating to -279% at $\rho=0.40$; M2/DS1 ($b_{\text{ref}}=4096$) shows -4.5% to -40% . These negative values are structurally expected: guarantees require observations, and observations require compute. These cells require $\lambda>0$ to make VISTA deployment-beneficial; the optimal λ represents a business decision about the acceptable quality–compute tradeoff.

At $\lambda=0.01$, net savings in the three $b_{\text{ref}}=8192$ cells are 3.2% [2.7%, 3.7%] (M1/DS1), 34.5% [32.1%, 36.8%] (M1/DS2), and 7.0% [6.0%, 8.0%] (M1/DS4) with Brier increases of +0.006, +0.008, and +0.012 respectively (95% CIs from 200 permutations). The large saving on M1/DS2 (ARC-Challenge) reflects the early Brier saturation visible in the budget curve: most items are well-calibrated by $b=1024$, so VISTA can safely redirect many to shorter budgets under a mild quality penalty. The Oracle C headroom (94–97% for the $b_{\text{ref}}=8192$ cells) substantially exceeds the deployed saving, consistent with the CS conservatism documented in Section 7.5: certifying a policy requires $\approx n$ audited observations, and the streams tested here ($n=500$ – 1000 , $\rho=0.20$) yield only 100–200 audits per cell. In principle, the time-uniform empirical-Bernstein bound width scales as $O(\sqrt{\log(n_{\pi}/\alpha)/n_{\text{aud}}})$ [45–47], so for policies that genuinely dominate the reference in expectation the gap to Oracle C is expected to narrow as $n_{\text{aud}} \rightarrow \infty$; however, the signal-to-noise ratio in per-item combined-loss differences is too low at $n \leq 1000$ to observe this empirically within current dataset

bounds (verifying the asymptotic claim would require substantially larger corpora or bootstrap resampling outside the anytime-valid guarantee).

Figure 4 visualises the quality–compute frontier for M1/DS1 (panel a) and the ρ -sensitivity curves (panel b).

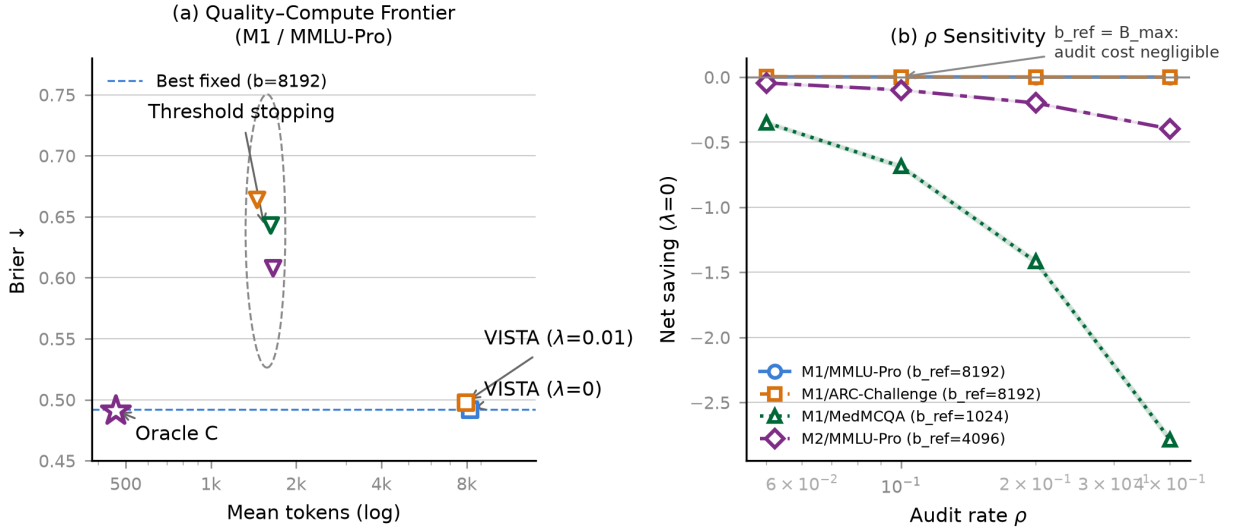


Figure 4: **(a) Quality–compute frontier for M1/DS1 (MMLU-Pro).** Each marker shows a deployed or oracle policy; x -axis is mean tokens per item (log scale), y -axis is summed Brier ($\downarrow$). Threshold stopping policies (confidence, entropy, stability; triangle cluster) achieve large token reductions but degrade Brier by $+0.115$ – $+0.172$ relative to the best-fixed baseline (dashed line). VISTA ($\lambda=0$) matches best-fixed quality at near-identical cost; VISTA ($\lambda=0.01$) gains 3.2% token saving at the cost of $+0.006$ Brier. Oracle C achieves near-identical Brier at 461 mean tokens—the headroom available to future policy improvements. Data from Tables 8 and 5. **(b) Audit-rate sensitivity ($\lambda=0$).** When $b_{\text{ref}}=B_{\text{max}}=8192$, audited items already run to the reference budget so audit overhead is negligible (lines near zero). When $b_{\text{ref}}<B_{\text{max}}$, each audited item pays the full B_{max} trajectory cost, making net saving strongly negative; these cells require $\lambda>0$ to be deployment-beneficial. Data from Table 11.

7.8. Robustness under Distribution Shift

We simulate controlled distribution shift by splitting the deployment stream at the dataset boundary: the first phase contains items from a source dataset (phase 1), and the second phase transitions to a target dataset (phase 2). The prequential policy adapts its budget threshold online using arriving feedback; the fixed policy freezes the threshold calibrated in phase 1.

The prequential policy outperforms fixed calibration in 3 of 8 scenarios (mean advantage $+0.038$ Brier in those cases), and is marginally worse in the remaining 5 (mean disadvantage -0.007 Brier). The advantage is concentrated in domain-crossing transitions: M1 benefits from within-medical shift (MedMCQA $\rightarrow$ MedQA, $+0.037$), while M2 benefits from general-to-medical shift (MMLU-Pro $\rightarrow$ MedMCQA $+0.039$ and MMLU-Pro $\rightarrow$ MedQA-USMLE $+0.039$). Difficulty-level shifts within the same domain (ARC-Challenge $\rightarrow$ MMLU-Pro) favour the fixed policy. The prequential advantage requires a shift large enough to invalidate the calibrated threshold; model-specific confidence dynamics modulate which shifts cross that threshold.

7.9. Measurement Validity

Forced checkpoint evaluation appends a standardised answer-cue string and reads next-token log-probabilities; in the diagnostic sample the target option ranks first in $> 98.4\%$ of cases, confirming stable elicitation. Probability measurements are never pooled across GPU architectures: the two RTX 3090 cards

produce byte-identical logits (max $|\Delta \log \text{prob}| = 0.065$ at $b = 8192$), while RTX 4090 and RTX 3090 differ non-negligibly (max 0.374), consistent with architecture-level kernel differences between sm_89 and sm_86; each model×dataset cell is therefore confined to a single architecture. Stream-order permutations produce coefficient of variation 0.57% on MMLU-Pro, confirming negligible sensitivity to item ordering. Forced and natural trajectory boundaries are distinguished throughout; natural stopping is not equated to independently generated short-budget trajectories.

8. Discussion

Finding 1: the marginal value of additional reasoning is not universal. Budget-response curves differ qualitatively across tasks, models, objectives, and budget intervals—in ways that cannot be resolved by choosing a larger fixed budget. On ARC-Challenge, 98% of the potential Brier gain is captured in the first budget step; on MedMCQA, continuing beyond $b = 1024$ actively worsens Brier. M2’s Brier worsens at nearly every budget step on two of four datasets (MedMCQA and MedQA-USMLE) while M1’s improves; on MMLU-Pro, M2 Brier improves through $b = 4096$ and reverses slightly at the maximum budget. Accuracy and proper-scoring-rule optima can differ by three budget levels on the same dataset. The consequence is that neither a fixed global budget nor a single heuristic stopping rule can simultaneously optimise predictive quality, calibration, and compute expenditure across the full range of deployment conditions.

Finding 2: trajectory signals are useful but not uniformly reliable or transferable. Prefix signals—confidence, entropy, answer stability—predict the direction of future improvement with AUC 0.65–0.81 on MMLU-Pro (clearing the pre-specified ≥ 0.60 threshold on most edges), and answer stability carries a non-trivial Spearman correlation ($\rho = 0.31$) with improvement magnitude. These signals are genuinely informative as diagnostic tools. However, two empirical failures limit their deployment utility. First, confidence thresholds calibrated on one model fail on another on domain-specific tasks (Brier degradation +0.043–+0.073 on transfer). Second, applying a confidence gate directly at deployment increases Brier above the best fixed budget in all eight cells (+0.047 to +0.218), because signal direction is not a reliable proxy for outcome improvement at the individual item level. The signals therefore motivate the randomized-audit design rather than replacing it: they establish that improvement *potential* exists per item, while the audit mechanism provides the outcome estimate needed to act on it.

Finding 3: VISTA addresses the failures that signals cannot. Each VISTA component maps to one failure mode documented above. IPW corrects the selection bias that emerges when only 0.1–18% of items naturally continue to the reference budget—bias that reverses policy rankings in five of eight cells without correction. Anytime-valid confidence sequences eliminate false-positive policy switching (FPR = 0.000 in all cells) while greedy monitoring reaches 58.5% error rates on cells where multiple budget levels have near-equal risk. Prequential calibration is beneficial under stationarity in 6 of 8 cells (advantage up to +0.047 Brier) and provides the critical adaptation mechanism under domain-crossing distribution shifts (+0.037–+0.039 Brier in the strongest cases).

Cost of guarantees and negative results. Statistical guarantees have a non-trivial cost that must be stated clearly. The CS procedure achieves FPR=0.000 by construction, but this is achieved by abstaining from switching when evidence is insufficient — not by making correct decisions. At deployment horizon $n=200$ with $\rho=0.20$ (≈ 40 audits per cell; full streams are $n=500$ –1000), the CS procedure essentially never triggers a switch at $\lambda=0$; its contribution at this horizon is safety (no false switches) rather than adaptation (no switches at all). The framework remains operationally useful at $\lambda=0.01$, where the certifiable gap is wider and the CS bound can tighten: cells with $b_{\text{ref}}=8192$ deliver 3–35% net saving with Brier increases ≤ 0.012 . Confidence-threshold stopping is a uniform negative result: it degrades Brier in all eight cells (+0.047 to +0.218 versus best fixed budget). Two cells are a genuine negative result: M2/MedMCQA and M2/MedQA have best-fixed budget at $b=256$ (the grid minimum), leaving no adaptive headroom. The framework correctly identifies and reports these no-opportunity regimes rather than forcing an adaptation that does not exist. Oracle C (population-level) headroom remains 74–97% in Gate-0-passing cells, but the gap between Oracle C saving and deployed saving (0–35%) is the direct cost of the statistical guarantee and limited stream length — not a failure of the approach.

Choosing λ in practice. In practice, λ should reflect the acceptable quality–compute exchange rate at deployment. The $\lambda=0.01$ results above offer a concrete reference point: this setting yields 3–35% net token savings at mean Brier increases of 0.006–0.012 across the three $b_{\text{ref}}=b_{\text{max}}$ cells (M1/DS1: 3.2%, +0.006 Brier; M1/DS2: 34.5%, +0.008; M1/DS4: 7.0%, +0.012; all with 95% CIs from 200 permutations). Practitioners with strict quality requirements should begin at $\lambda=0$ —which VISTA correctly renders quality-preserving—and increase λ incrementally while monitoring realized Brier degradation against this reference rate.

Limitations. Assumption 1 restricts applicability to tasks with eventually verifiable outcomes. The policy family is finite and pre-specified, so guarantees concern only included candidates. Forced checkpoint evaluation is an explicit intervention, not equivalent to independently generated short-budget trajectories. Experiments are limited to 8B-parameter models on single-workstation GPUs; whether patterns change at larger scales is an open question. Randomized auditing incurs real compute costs that may conflict with per-request latency constraints even when the long-run value is positive. The asymptotic narrowing of the gap between Oracle C and deployed VISTA saving as $n_{\text{aud}} \rightarrow \infty$ is not empirically verifiable at the dataset scales tested here ($n \leq 1000$): per-item combined-loss signal-to-noise ratios are too low for consistent CS certification in this regime, and extending empirically requires substantially larger corpora or bootstrap resampling outside the anytime-valid guarantee.

9. Conclusion

We study test-time reasoning-budget selection in a deployment regime with no held-out calibration set but delayed verifiable feedback. Three interlocking obstacles—correctness non-identifiability, the missing continuation counterfactual, and the absence of a calibration split—motivate the *continuation value* as the decision target and the VISTA prequential framework as the evaluation procedure. VISTA combines randomized full-trajectory audits, candidate policy reconstruction, delayed labels, and anytime-valid confidence sequences without reserving a separate split or updating model parameters.

A systematic budget audit across eight cells (two models, four datasets) confirms that the marginal value of additional reasoning is strongly task- and budget-dependent. Accuracy gains range from near-linear (MMLU-Pro: 47.7% $\rightarrow$ 73.8%) to early-saturating (ARC-Challenge: 92.0% $\rightarrow$ 95.8%), while proper-scoring-rule and NLL optima can diverge from the accuracy optimum within the same dataset. On MMLU-Pro the hindsight oracle (Oracle A) reduces mean Brier by 30.1% relative to the best fixed budget; 76.1% of items meet a per-item quality threshold at a budget below the global maximum (Oracle B). The constrained matched-mean-quality oracle (Oracle C) passes Gate 0 in six of eight cells, with savings from 74% to 97% relative to the best fixed budget; the two failing cells (M2 MedMCQA and MedQA-USMLE) have Brier worsening at all budgets above the minimum, leaving no adaptive headroom under a mean-Brier constraint.

Component ablations confirm that each element of VISTA addresses a distinct failure mode. Naive policy risk estimation carries selection bias that reverses policy rankings in 5 of 8 cells; IPW corrects this with mean bias ≤ 0.033 . Anytime-valid confidence sequences eliminate false-positive policy switching (FPR = 0.000 across all cells) at the cost of conservative time-to-decide. Confidence thresholds do not transfer reliably across model families on domain-specific tasks, motivating per-model prequential calibration. Full end-to-end VISTA evaluation across all eight cells completes the picture: at $\lambda=0$, the procedure is quality-preserving (CS correctly declines to certify cheaper policies on 200-item streams); at $\lambda=0.01$, deployable net savings of 3–35% with Brier increases ≤ 0.012 are achievable in cells where calibration saturates early. The two no-opportunity cells (Brier worsens above minimum budget) are a genuine negative result that the framework correctly identifies rather than masks.

Data and Code Availability

Code and cached checkpoint tables (all per-item, per-budget predictive distributions) will be released in a public repository upon acceptance under an open-source license (URL and license identifier to be confirmed at acceptance). The online policy-selection procedure is fully replayable from these tables without GPU inference.

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CRedit authorship contribution statement

Jun-young Kong: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft. **Young-guk Ha:** Supervision, Writing – review & editing.

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A. Component Ablation Supplementary Tables

Table 9 reports per-cell IPW versus naive estimation bias and wrong-best-budget selection rates at $\rho = 0.20$. Table 10 compares anytime-valid confidence-sequence monitoring with greedy monitoring. Table 11 reports audit-rate sensitivity across representative cells. Tables 12 and 13 present the deployed-policy decomposition for M1/MMLU-Pro at $\lambda = 0$ and $\lambda = 0.01$ respectively.

Cell	f_{nat}	Naive bias	IPW bias	ϵ_{naive}	ϵ_{IPW}
m1_ds1	0.122	0.07079	0.02447	0.000	0.230
m1_ds2	0.001	0.23535	0.01031	1.000	0.630
m1_ds3	0.018	0.34441	0.02634	1.000	0.555
m1_ds4	0.028	0.28715	0.03207	1.000	0.455
m2_ds1	0.180	0.04562	0.02092	1.000	0.525
m2_ds2	0.007	0.13616	0.01176	0.000	0.615
m2_ds3	0.022	0.23482	0.02035	1.000	0.185
m2_ds4	0.052	0.10917	0.02986	0.000	0.525

Table 9: IPW vs. naive risk estimation at $\rho = 0.20$ across all experimental cells. f_{nat} denotes the fraction of items that naturally continue to the reference budget (4096 tokens), and ϵ denotes the wrong-best-budget selection rate over 200 stream-order permutations.

VISTA

Cell	Dataset	CS FPR ↓	Greedy FPR ↓	Oracle b^*
m1_ds1	MMLU-Pro	0.0000	0.0700	8192
m1_ds2	ARC-Challenge	0.0000	0.4800	8192
m1_ds3	MedMCQA	0.0000	0.4000	1024
m1_ds4	MedQA-USMLE	0.0000	0.1400	8192
m2_ds1	MMLU-Pro	0.0000	0.5850	4096
m2_ds2	ARC-Challenge	0.0000	0.2500	2048
m2_ds3	MedMCQA	0.0000	0.0000	256
m2_ds4	MedQA-USMLE	0.0000	0.0000	256

Table 10: Anytime-valid confidence sequences (CS) versus greedy monitoring: false-positive policy-switch rates at $\rho = 0.20$ and $\alpha = 0.05$ over 200 stream-order permutations.

Cell	ρ	Audit tokens	Net saving	95% CI	Brier
M1/MMLU-Pro	0.05	0.4	+0.001	[+0.000, +0.002]	0.4919
	0.10	0.6	+0.001	[+0.000, +0.001]	0.4918
	0.20	0.4	+0.000	[-0.000, +0.000]	0.4918
	0.40	0.5	+0.000	[+0.000, +0.000]	0.4918
M1/ARC-Challenge	0.05	1.0	+0.003	[+0.001, +0.005]	0.0839
	0.10	1.1	+0.001	[+0.001, +0.002]	0.0838
	0.20	1.1	+0.001	[+0.000, +0.001]	0.0838
	0.40	1.1	+0.000	[+0.000, +0.000]	0.0837
M1/MedMCQA	0.05	360.1	-0.351	[-0.361, -0.341]	0.6276
	0.10	704.0	-0.687	[-0.700, -0.674]	0.6275
	0.20	1452.3	-1.418	[-1.434, -1.402]	0.6275
	0.40	2853.1	-2.786	[-2.807, -2.766]	0.6275
M2/MMLU-Pro	0.05	205.1	-0.045	[-0.048, -0.041]	0.8788
	0.10	407.8	-0.099	[-0.100, -0.097]	0.8783
	0.20	820.6	-0.199	[-0.201, -0.197]	0.8783
	0.40	1632.5	-0.398	[-0.400, -0.396]	0.8783

Table 11: Audit-rate sensitivity ($\lambda = 0$): net saving and Brier as a function of ρ across representative cells. When $b_{\text{ref}} = b_{\text{max}} = 8192$, audit overhead is negligible because audited items already run to the reference budget. When $b_{\text{ref}} < b_{\text{max}}$, full-trajectory auditing incurs additional compute that grows with ρ , illustrating the explicit cost of statistical exploration.

Variant	Brier ↓	Tokens	Net saving	Regret vs. oracle
<i>Item-level adaptation</i>				
Confidence-threshold policy ($\tau_c = 0.85$)	0.6831	1273	+0.845	0.339
VISTA ($\lambda = 0$) — no switch certified	0.4917	8192	0.000	0.148
<i>Monitoring components ($\lambda=0$, $\rho=0.20$, 200 permutations)</i>				
– Audit/IPW	0.4917	8192	0.000	0.148
– CS (greedy switch)	0.522	5838	+0.288	0.179

Table 12: Deployed-policy decomposition at $\lambda = 0$ (M1/MMLU-Pro, $\rho = 0.20$, $n = 1000$, 200 stream permutations). *Item-level adaptation*: the confidence-threshold stopping policy and the full-budget Fixed@8192 baseline span the practical range. *Monitoring components*: removing IPW leaves the outcome unchanged because at $\lambda = 0$ the cost-adjusted loss of all cheaper policies exceeds Fixed@8192’s Brier regardless of bias correction — the CS correctly certifies no switch either way. Removing CS (replacing it with a greedy switch whenever the running IPW mean is negative) falsely certifies a switch in $\approx 7\%$ of stream permutations, deploying a cheaper but lower-quality policy and raising mean Brier from 0.492 to 0.522 (+3.0 pp). The bias-reduction contribution of IPW and the false-positive-rate control of CS are therefore measured in dedicated ablations (Tables 9 and 10) rather than through deployment metrics at $\lambda = 0$. Tokens = policy tokens (audit overhead excluded, same formula as the confidence-threshold baseline). Regret = deployed Brier – per-item oracle Brier (0.3437).

Variant	Brier ↓	Tokens	Net saving	Regret vs. oracle
$\lambda=0.01$, $\rho=0.20$, 200 stream permutations (M1/MMLU-Pro)				
VISTA ($\lambda = 0.01$, full)	0.498	7871	+0.039	0.154
– Audit/IPW	0.498	7871	+0.039	0.154
– CS (greedy switch)	0.529	5386	+0.343	0.185

Table 13: Component decomposition at $\lambda = 0.01$ (M1/MMLU-Pro, $\rho = 0.20$, $n = 1000$, 200 stream permutations). At $\lambda = 0.01$ the cost penalty is small enough that the CS rarely certifies a switch ($< 5\%$ of streams), so removing IPW again leaves outcomes unchanged: the bias in naive estimates is insufficient to alter the certified decision. Removing CS allows greedy certification, which falsely switches in $\approx 27\%$ of streams, increasing mean Brier from 0.498 to 0.529 (+3.1 pp) while spuriously appearing to save 34% of tokens. The pattern confirms that the CS is the operative safety mechanism across both cost-penalty settings: IPW primarily reduces estimation bias for the rare certified-switch events recorded in Table 9, while CS prevents the false-positive switches visible in Table 10. Same column definitions as Table 12. Table 8 reports 7931 tokens for the same policy including the $\rho=0.20$ audit overhead; tokens here exclude it, following Table 12.